freight rate prediction report

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validation and split approach

the training data covers january to october 2025 and the validation set is november and december, so the real task is predicting two unseen future months. i split the development data by time to copy that situation: train on january to august, test on september and october.

a random split would put loads from the same days on both sides, so the model would only fill gaps between dates it already saw and the score would look better than it really is.

about 670 loads (1.4%) have a rate per mile far above or below normal and none of the features explain them (distance, equipment, weight, date and route all look normal). i removed them from training only and kept them in the test, because the real validation set will contain loads like these.

everything learned from data (the weight median for missing values, the equipment encoder and the lane encoder) is fitted on the training part only. after choosing the model i retrained it on all of january to october, because september and october are the months closest to what we predict.

data cleaning

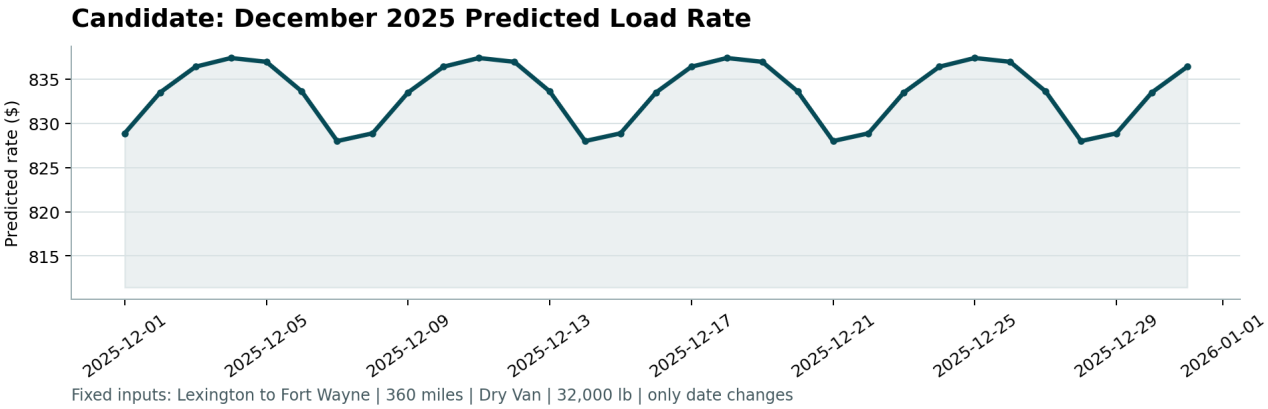
- negative weights (292 in training, 145 in validation) closely mirror the positive weights, so they are a sign error, fixed with abs()
- missing weights (300 in training, 165 in validation) are filled with the training median
- dropped quote_signal: target leakage, for many loads it equals the rate per mile, and it is missing from the december file
- dropped market_index: no signal (correlation 0.03), missing values, and missing from the december file
- dropped load_id, and replaced the city names with their coordinates (every city has exactly one set). december has no coordinates so they are looked up from training
- no duplicate load ids and no duplicate rows

results on september and october

model	MAE
baseline (median rate per mile x distance)	256.94
random forest	106.45
random forest + lane feature	105.67
lightgbm + lane feature	107.84
average of random forest and lightgbm (final)	**102.45**

final model R2: 0.83. the lane feature is the typical rate per mile of each pickup and delivery pair, computed with an out of fold target encoder so a load never sees its own price.

december prediction chart



the chart shows a weekly pattern: wednesday and thursday highest, the weekend lowest, the same pattern as in the training data. the swing is small, about 828 to 837 dollars (around 1%).

christmas looks like a normal thursday. this is a limitation, not a bug: the training data ends in october, and tree models cannot predict outside the range they were trained on, so december is treated like october and no holiday effect can be learned. data from previous decembers would be needed for that.